



Right Triangle Trigonometry

Right triangles can be solved using trigonometric ratios and the Pythagorean theorem.

Using Similarity in Right Triangles

Definition

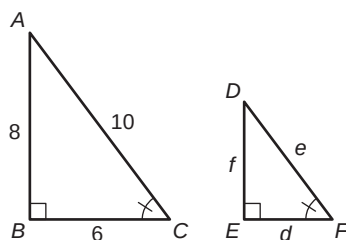
Two triangles are **similar** if their corresponding angles are congruent and their corresponding side lengths are proportional.

Because the lengths of corresponding sides of similar triangles are proportional, the ratios of corresponding side lengths are equal.

► **Think About It** When making statements about similarity and congruence, always name figures according to their corresponding parts.

EXAMPLE 1

What is the value of the ratio $\frac{f}{e}$ in $\triangle DEF$?



► **Remember** To indicate the length of a segment, use the names of its endpoints without a segment symbol over them. For example, AB indicates the length of $\overline{AB}$.

SOLUTION

$\triangle ABC$ and $\triangle DEF$ are right triangles with $\angle B \cong \angle E$ and $\angle C \cong \angle F$.

Because two angles of $\triangle ABC$ are congruent to two angles of $\triangle DEF$, the third pair of angles, $\angle A$ and $\angle D$, must also be congruent.

Therefore, the triangles are similar, with $\overline{AB}$ corresponding to $\overline{DE}$ and $\overline{AC}$ corresponding to $\overline{DF}$, which means that $\frac{AB}{AC} = \frac{DE}{DF} = \frac{f}{e} = \frac{8}{10} = \frac{4}{5}$. ■

Determining Trigonometric Ratios in Right Triangles

The ratios of side lengths in right triangles are properties of the angles of the triangle. They are called trigonometric ratios and have specific names.

Definitions

For any acute angle of a right triangle,

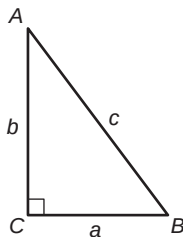
- The **sine (sin)** of an angle is the ratio of the length of the leg opposite the angle to the length of the hypotenuse.
- The **cosine (cos)** of an angle is the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse.
- The **tangent (tan)** of an angle is the ratio of the length of the leg opposite the angle to the length of the leg that is adjacent to the angle.

In $\triangle ABC$,

$$\bullet \sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{c}$$

$$\bullet \cos A = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{b}{c}$$

$$\bullet \tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{a}{b}$$



► **Think About It** The hypotenuse will always be the side opposite the right angle. But which sides are adjacent and opposite depends on which acute angle is referenced.

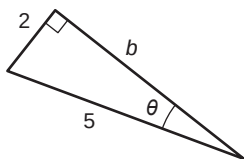
EXAMPLE 2

In a right triangle, θ is an acute angle and $\sin \theta = \frac{2}{5}$. Determine $\cos \theta$ and $\tan \theta$.

SOLUTION

Step 1 Draw the triangle and choose an acute angle to label θ . Since

$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$, label the opposite side with length 2, and the hypotenuse with length 5.



Step 2 To calculate $\cos \theta$ and $\tan \theta$, you must first determine the missing side length, b . Use the Pythagorean theorem, with $a = 2$ and $c = 5$.

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 2^2 + b^2 &= 5^2 \\ 4 + b^2 &= 25 \\ b^2 &= 21 \\ b &= \sqrt{21} \end{aligned}$$

- **Think About It** Because lengths can't be negative, always discard negative side length values that result from solving equations such as $b^2 = 21$.

Step 3 Use the definitions to find the unknown ratios.

$$\begin{aligned}\cos A &= \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\sqrt{21}}{5} \\ \tan A &= \frac{\text{opposite}}{\text{adjacent}} = \frac{2}{\sqrt{21}} = \frac{2\sqrt{21}}{\sqrt{21} \cdot \sqrt{21}} = \frac{2\sqrt{21}}{21} \quad \blacksquare\end{aligned}$$

Applying Trigonometric Ratios

You can use trigonometric ratios to solve for angle measures and side lengths in a variety of situations. After you read a problem, draw and label a triangle that models the scenario. Then set up an equation involving the appropriate trigonometric ratio and solve for the unknown measure.

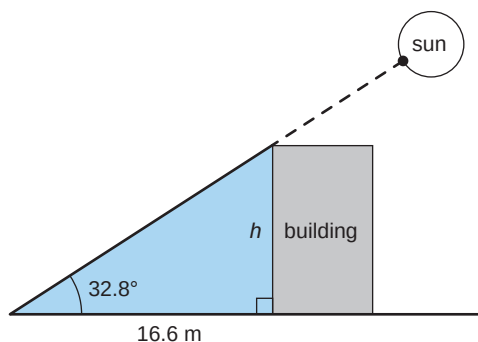
EXAMPLE 3

The shadow of a vertical building is 16.6 m long when the angle of elevation of the sun is 32.8° . What is the height of the building to the nearest tenth of a meter?

- **Think About It** The angle of elevation of an object is the upward angle that the object makes with the horizontal. The angle of depression of an object is the downward angle that the object makes with the horizontal. The angle of depression from Object A to Object B is the same as the angle of elevation from Object B to Object A.

SOLUTION

Draw a right triangle that models the situation. Label the unknown height as h .



The angle of elevation, 32.8° , is a known value. The unknown height, h , is opposite that angle. The shadow length, 16.6 m, is the length of the side adjacent to the angle. Use the tangent since it is the ratio that involves the opposite and adjacent side lengths.

$$\begin{aligned}\tan(32.8^\circ) &= \frac{h}{16.6} \\ \tan(32.8^\circ) \cdot 16.6 &= h \\ 10.7 &\approx h\end{aligned}$$

The building is approximately 10.7 m high. ■